

Supplementary Material
for
**A Cloud-Based Two-Layer Text Classification Framework for Mental Health
Screening with Sarcasm and Emoji-Aware Sentiment Analysis**

This Supplementary Information file contains:

- ***Supplementary Note 1***
Synthetic Dataset Construction: Design Rationale and Category Definitions
- ***Supplementary Table S1***
Representative Synthetic Text Samples by Mental Health Category
- ***Supplementary Table S2***
Extended Per-Class Classification Performance Metrics
- ***Supplementary Table S3***
Examples of Sarcastic and Emoji-Augmented Text Samples with Classification Outcomes
- ***Supplementary Methods***
Azure Language Studio: Configuration and Training Protocol
- ***Supplementary Note 2***
Clinical Annotation Protocol for External Validation Dataset
- ***Supplementary Table S4***
Confusion Matrix Values from Clinically Reviewed Validation Dataset (125 Samples)
- ***Supplementary Discussion***
Deployment Considerations and Limitations for Real-World Applications

Supplementary Note 1: Synthetic Dataset Construction: Design Rationale and Category Definitions

The primary training and testing dataset used in this study was developed entirely through systematic manual curation of synthetic text samples. This note describes the design rationale, category definitions, and linguistic diversity strategies employed during dataset development.

1.1 Rationale for Synthetic Data Development

The decision to develop a fully synthetic dataset was driven by two primary considerations: (i) the ethical imperative to avoid collecting, storing, or redistributing sensitive mental health content generated by real individuals, and (ii) the absence of publicly available clinically annotated datasets that provide labels corresponding to all five target categories considered in this study. The use of synthetic data ensured complete compliance with privacy requirements while allowing precise control over linguistic characteristics, category balance, and stylistic diversity.

1.2 Observational Reference Analysis

Prior to dataset construction, an observational analysis of publicly accessible mental health discussion forums and online communities was conducted. This analysis was limited strictly to identifying recurring linguistic patterns, emotional expression styles, communication conventions, and thematic characteristics associated with each mental health category. No user-generated content was copied, downloaded, stored, or reproduced at any stage. The analysis informed the vocabulary, phrasing conventions, and structural patterns incorporated into the synthetic samples.

1.3 Category Definitions and Linguistic Scope

Each of the five classification categories was operationally defined in terms of the linguistic and thematic patterns targeted during sample generation:

- a) **Anxiety:** Samples in this category were designed to reflect linguistic patterns associated with excessive worry, anticipatory fear, physiological tension cues (e.g., racing heart, inability to breathe), rumination, avoidance behaviors, and difficulty managing uncertainty. Both direct and indirect expressions of anxious states were included.
- b) **Depression:** Samples targeted linguistic markers of persistent low mood, loss of interest or motivation, fatigue, hopelessness, emptiness, cognitive slowing, and withdrawal. The category included both explicit expressions of depressive affect and more subtle, indirect formulations consistent with anhedonia and low energy.
- c) **Post-Traumatic Stress Disorder (PTSD):** Samples were constructed to reflect linguistic patterns associated with trauma-related re-experiencing (e.g., intrusive memories, flashbacks), hypervigilance, emotional numbing, avoidance of trauma-related cues, and disrupted sense of safety. References to persistent nightmares and difficulty trusting others were also represented.
- d) **Social Anxiety Disorder (SAD):** Samples reflected linguistic patterns associated with fear of negative evaluation, avoidance of social situations, self-consciousness, anticipatory anxiety before social encounters, and perceived social incompetence. Expressions of embarrassment, blushing, and reluctance to engage in public activities were incorporated.
- e) **Suicidal Ideation and Behavior:** Samples included linguistic patterns associated with passive death wishes, active suicidal ideation, expressions of hopelessness and perceived burdensomeness, feelings of entrapment, and statements indicative of low perceived reasons for living. All samples in this category were synthetic and reviewed to ensure no real crisis content was reproduced.

1.4 Linguistic Diversity Strategies

The dataset was intentionally designed to encompass multiple communicative registers and linguistic styles, including:

- Direct emotional disclosures (explicit statements of emotional states);
- Indirect emotional expressions (oblique references to distress without explicit labeling);
- Ambiguous or underspecified statements (expressions that suggest distress without definitive disclosure);
- Sarcastic expressions (statements where surface-level positive language contradicts the implied emotional context);
- Emoji-based communication (samples incorporating one or more emotionally expressive emojis that contribute to or modify the communicative intent);
- Informal and abbreviated language (social media writing styles including abbreviations, misspellings, and non-standard punctuation);
- Varying levels of linguistic complexity (from simple, short statements to more elaborate, multi-clause expressions).

1.5 Dataset Composition and Split

The final dataset consisted of 500 synthetic text samples distributed equally across the five target categories (100 samples per category). An 80% training and 20% testing split was applied, resulting in 400 training samples and 100 testing samples. All samples were preprocessed through a standardized pipeline including text normalization, lowercasing, punctuation normalization, token standardization, and formatting cleanup prior to ingestion into Azure Language Studio.

Supplementary Table S1: Representative Synthetic Text Samples by Mental Health Category

Table S1 presents representative synthetic text samples from each of the five target categories. Samples illustrate the range of communicative styles included in the training and evaluation dataset, including direct disclosures, indirect expressions, sarcastic statements, and emoji-augmented text. All samples are fully synthetic and were generated during the dataset construction process described in Supplementary Note 1.

Category	Linguistic Style	Representative Synthetic Sample
Anxiety	Direct	I haven't slept properly in weeks. Every night I lie awake running through worst-case scenarios I know will probably never happen, but I can't stop.
Anxiety	Indirect	Lately it feels like something bad is always just around the corner. I keep checking things twice, three times, just to be sure.
Anxiety	Sarcastic	Oh great, another presentation at work. Nothing like standing in front of everyone while your heart tries to escape your chest 😊
Depression	Direct	I don't enjoy anything anymore. Even things I used to love feel pointless. Getting out of bed feels like climbing a mountain.
Depression	Indirect	Everything just feels... grey. I'm not sad exactly, just empty. Like someone turned the volume all the way down.

Category	Linguistic Style	Representative Synthetic Sample
Depression	Emoji-based	Another day, another nothing. 🌀😬 Went through the motions. Came home. Repeat.
PTSD	Direct	I had the dream again last night. The same one. I woke up shaking and couldn't remember where I was for a few seconds.
PTSD	Indirect	I can't go back to that part of town. I know it's been years but certain sounds just take me right back and my whole body reacts before I can think.
PTSD	Sarcastic	Really love how my brain decided that car backfiring = full panic mode. Very useful, very proportionate, totally normal 😬
Social Anxiety Disorder	Direct	I turned down another party invite. I know they'll think I'm rude but the thought of standing there not knowing what to say to people makes me feel sick.
Social Anxiety Disorder	Indirect	I rehearse conversations before I have them. Every one. Then replay them after to work out what I did wrong.
Social Anxiety Disorder	Emoji-based	Said the wrong thing in the meeting today 😬 I've been replaying it for three hours. Everyone definitely noticed.
Suicidal Ideation	Direct	Sometimes I think about how much easier it would be if I just wasn't here anymore. I'm so tired of fighting to feel okay.

Category	Linguistic Style	Representative Synthetic Sample
Suicidal Ideation	Indirect	I keep imagining what it would be like for everyone if things were just... different. If I wasn't a problem for anyone anymore.
Suicidal Ideation	Sarcastic	Oh don't worry about me, I'm totally fine. Just thinking again about how everyone's life would be simpler without me around. No big deal 😊

Note: All samples presented are synthetic and were generated for research purposes only. No real user-generated content is represented. Category labels are as defined in the main manuscript.

Supplementary Table S2: Extended Per-Class Classification Performance Metrics

Table S2 provides extended per-class classification metrics from evaluation of the Azure Custom Text Classification model on the held-out testing dataset (100 samples; 20 per category). Values are derived from an 80/20 training-testing split applied to the primary synthetic dataset.

CATEGORY	PRECISION (%)	RECALL (%)	F1 SCORE
<i>Anxiety</i>	95.24	100.00	0.976
<i>Depression</i>	100.00	100.00	1.000
<i>PTSD</i>	100.00	95.00	0.974
<i>Social Anxiety Disorder</i>	90.48	100.00	0.950
<i>Suicidal Ideation and Behavior</i>	100.00	90.00	0.947
OVERALL (MACRO-AVG)	96.97	96.97	0.970

Supplementary Table S3: Sarcastic and Emoji-Augmented Text Samples with Classification

Outcomes

Table S3 presents representative examples from the qualitative evaluation of sarcastic expressions and emoji-augmented text. For each example, the original sarcastic statement and its emoji-augmented counterpart are shown alongside the predicted category and the model's confidence score for each version. Confidence scores represent the probability assigned by the Azure Custom Text Classification service to the predicted category. All samples are synthetic.

Original Text	Emoji-Augmented Version	Predicted Category	Conf. Without Emoji	Conf. With Emoji
Yeah I'm totally fine, just haven't eaten in two days and haven't left the room.	Yeah I'm totally fine, just haven't eaten in two days and haven't left the room 	Depression	0.86	0.93
Another social event I managed to get out of. The relief is indescribable.	Another social event I managed to get out of. The relief is indescribable 	Social Anxiety Disorder	0.82	0.91
Certain smells take me back immediately. I hate that I have no control over it.	Certain smells take me back immediately. I hate that I have no control over it 	PTSD	0.89	0.95
It's fine, I'll just lie here staring at the ceiling like every other night.	It's fine, I'll just lie here staring at the ceiling like every other night 	Anxiety	0.78	0.88

Original Text	Emoji-Augmented Version	Predicted Category	Conf. Without Emoji	Conf. With Emoji
Happiness must have forgotten my address	Happiness must have forgotten my address 🙄	Suicidal Ideation and Behavior	0.97	0.99

Conf. = Confidence score (0–1 scale) assigned by Azure Custom Text Classification to the predicted category. In all presented examples, emoji augmentation preserved the predicted category while increasing classification confidence, consistent with the findings reported in the main manuscript.

Supplementary Methods: Azure Language Studio Configuration and Training Protocol

- ***Azure Custom Text Classification Setup***

A Custom Text Classification project was created in Azure Language Studio and configured as a single-label multi-class classification task. Synthetic text samples were uploaded through Azure Blob Storage and manually assigned to one of five target categories: Anxiety, Depression, PTSD, Social Anxiety Disorder, and Suicidal Ideation and Behavior. The dataset was partitioned using an 80% training and 20% testing split within Azure Language Studio.

- ***Model Training***

Training was performed using the standard Azure Language Studio training pipeline. Azure manages feature extraction, representation learning and model selection internally; therefore, architectural details such as embedding dimensions, optimizer settings, learning rates, and training schedules are not exposed to users. Model performance was evaluated using the platform-generated Precision, Recall, and F1-score metrics on the held-out test set.

- ***Deployment and API Integration***

Following training, the model was deployed as a REST API endpoint within Azure Language Services. The Android application communicated with the endpoint using the OkHttpClient library over encrypted HTTPS connections. Classification requests were transmitted in JSON format and authenticated using Azure subscription keys.

- ***Text Preprocessing***

Prior to classification, input text underwent lightweight preprocessing consisting of Unicode normalization, whitespace normalization, punctuation normalization, and line-break standardization. Emoji characters were preserved to retain potentially relevant contextual information during classification.

Supplementary Note 2: Clinical Annotation Protocol for External Validation Dataset

This note describes the annotation protocol used for the independent clinical review of the external validation dataset. This dataset was not used during model training and was employed exclusively for secondary evaluation, as described in the main manuscript.

Annotator Qualifications

The external validation dataset was independently reviewed and annotated by qualified mental health professionals, including practitioners with postgraduate training in psychiatry and clinical psychology. Annotators were selected on the basis of their clinical expertise in the assessment and diagnosis of anxiety disorders, depressive disorders, PTSD, and related conditions. All annotators provided informed consent to participate in the annotation process.

Annotation Materials

Each annotator received: (i) a set of 125 synthetic text samples (25 per category), presented in a randomized order without category labels or grouping information; (ii) an annotation guideline document defining the five target categories in clinical terms and providing example linguistic markers associated with each; and (iii) a structured annotation form for recording category assignments and optional free-text observations regarding sample ambiguity or boundary cases.

Category Definitions Provided to Annotators

Annotators were provided with the following operational definitions, derived from standard diagnostic frameworks:

- a) **Anxiety:** Text reflecting persistent and excessive worry, fear, or apprehension that the writer reports as difficult to control. May include references to physiological arousal, avoidance behaviors, or rumination.
- b) **Depression:** Text reflecting persistent low mood, loss of interest or pleasure, fatigue, hopelessness, or cognitive difficulties. May include references to social withdrawal, emptiness, or reduced motivation.

- c) **PTSD:** Text reflecting trauma-related re-experiencing (intrusive memories, nightmares, flashbacks), hypervigilance, emotional numbing, or avoidance of trauma-related stimuli.
- d) **Social Anxiety Disorder:** Text reflecting marked fear or anxiety about social situations, concern about negative evaluation by others, avoidance of social interactions, or significant distress associated with anticipated or actual social contact.
- e) **Suicidal Ideation and Behavior:** Text reflecting passive death wishes, active thoughts of suicide, expressed hopelessness, perceived burdensomeness, or statements suggestive of low reasons for living.

Conflict Resolution

In cases where annotator assignments differed from the intended category label, discrepancies were reviewed by a senior mental health professional who examined the linguistic content of the sample and provided a final adjudicated label. All adjudicated labels were used in the final validation confusion matrix presented in the main manuscript. Inter-annotator agreement was not formally computed because the primary purpose of the annotation exercise was validation against model predictions rather than establishing a gold standard from multiple annotators.

Ethics and Confidentiality

All annotators were informed that the samples were entirely synthetic and contained no real patient information. Annotators were instructed to treat samples as if they were real clinical communications for the purpose of the annotation exercise. Annotation data was handled confidentially and used solely for the purpose of evaluating model performance.

Supplementary Table S4: Confusion Matrix – Clinically Reviewed Validation Dataset (125 Samples)

Table S4 presents the full numerical confusion matrix generated from evaluation of the Azure Custom Text Classification model on the independently annotated validation dataset (N = 125; 25 samples per category). Rows represent actual (annotated) categories; columns represent model-predicted categories. Diagonal cells (shaded green) indicate correct classifications. Off-diagonal values indicate misclassifications. SAD = Social Anxiety Disorder.

Actual\ Predicted	Anxiety	Depression	PTSD	SAD	Suicidal Ideation
Anxiety	23	0	0	2	0
Depression	0	23	0	0	2
PTSD	0	0	25	0	0
SAD	2	0	0	23	0
Suicidal Ideation	0	0	0	0	25

Note: Misclassifications were confined to semantically related category pairs: Anxiety ↔ Social Anxiety Disorder (2 samples each direction) and Depression → Suicidal Ideation and Behavior (2 samples). These patterns are consistent with known linguistic overlap between conceptually proximal mental health constructs.

Supplementary Discussion: Deployment Considerations and Limitations for Real-World Applications

Deployment and Scalability

The proposed framework utilizes Azure Language Services and Firebase Realtime Database to support cloud-based text classification, data storage, and real-time synchronization. The architecture supports scalable deployment and asynchronous processing, although response latency may vary depending on network conditions and cloud resource availability.

Privacy and Data Governance

User-submitted text is transmitted to Azure services for analysis and stored within Firebase under authenticated user accounts. Access to stored information is restricted to authorized administrators. Organizations deploying the system should implement appropriate consent procedures, access controls, and data retention policies in accordance with applicable privacy regulations.

Intended Use

The framework is intended as a text classification and screening support tool and should not be interpreted as a clinical diagnostic system. Classification outputs are designed to identify linguistic patterns associated with mental health-related discourse and may assist in directing users toward appropriate support resources.

Limitations

The study relies primarily on synthetic datasets for model development and evaluation. Although an additional clinically reviewed validation dataset is being developed, further evaluation on real-world datasets is necessary to establish broader generalizability. The current implementation is also limited to English-language text.